Per-Schema Fee Markets for Attestation-Native Chains

EIP-1559-Style Per-Schema Base Fees with PoUA-Coupled Burn

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Abstract

A unified fee market across all transactions assumes homogeneous demand. Attestation-native chains break that assumption: a high-throughput schema (AI-provenance receipts at millions of attestations per day) and a low-throughput, high-value schema (sovereign-identity proofs at hundreds per day) have fundamentally different demand profiles, fee elasticities, and inclusion preferences. A single base-fee curve cannot fit both regimes without penalizing one to serve the other.

This paper proposes **per-schema fee markets** as a runtime primitive on Ligate Chain: each registered schema σ carries its own fee-market state $(b_\sigma, u_\sigma, T_\sigma)$, with EIP-1559-style base-fee adjustment, a per-schema tip floor, and a configurable burn-versus-routing split. The mechanism is composable with **sponsored-gas patterns** (Iris MCP relayer paying fees on behalf of autonomous agents) and **native delegation** (companion paper): signer and fee-payer are independent fields on the same transaction, and authorization checks for each compose orthogonally.

Three contributions. First, we specify the protocol-level mechanism (§4): the per-schema base-fee adjustment $b_\sigma(t+1) = b_\sigma(t) \cdot (1+\xi \cdot (u_\sigma - T_\sigma)/T_\sigma)$, the tip auction within schema capacity, the burn split $\text{burn}(\sigma) = \tau_{\text{burn}} + (1-\tau_{\text{burn}})(1-\rho_\sigma)$, and the integration with PoUA's adaptive τ_{burn} rebase. Second, we prove that the per-schema isolation preserves **PoUA Lemma 1's cost-to-grind floor** even at the most aggressive routing ($\rho_\sigma = 0.5$ for every schema): the burned fraction per attestation remains $\geq \tau_{\text{burn}} \cdot 0.5 \cdot b_\sigma$, and the chain-level cost-to-grind formula $F_{\text{net}} \geq \tau_{\text{burn}} \cdot \Delta r/(\eta \cdot \alpha_{\text{eff}})$ holds per-schema with the same constants. Third, we recommend a v0 calibration $(\xi, T_\sigma, \rho_\sigma) = (1/8, 0.5, 0)$ with profile-specific deviations for bursty, latency-critical, and high-value-low-volume schemas (§4.2 calibration table).

The mechanism's central design choice is to make the **schema** the fee-market unit. The chain already accounts for attestation work per-schema (PoUA §4.3 reputation update is schema-tagged). Adding per-schema fee state is a natural extension: state already exists, demand profiles differ by orders of magnitude per-schema, and per-schema isolation means a high-utilization schema cannot drag down the base fee of an unrelated schema. The §6 comparison positions per-schema fees against EIP-1559, Solana priority fees, Cosmos fee market module, and Aptos gas pricing; the §7 implementation walkthrough connects to ligate-chain's Sovereign SDK integration; the §5 security analysis bounds four cross-schema attack patterns.

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1. Introduction

1.1 The Heterogeneous-Demand Thesis

Blockchain fee market design has, until recently, assumed a single workload. Ethereum, the canonical example, fits this assumption: token transfers, DeFi interactions, and contract calls share enough demand-curve shape that a single base fee with global adjustment dynamics serves them adequately. EIP-1559's unified base fee adjusts to chain-wide congestion. The cost of mixing workloads is paid by everyone uniformly, which is acceptable when no workload is structurally different from the others.

Attestation-native chains break that assumption decisively. Consider two schemas that might coexist on Ligate Chain at maturity. **Themisra `proof-of-prompt/v1`** is the AI-provenance receipt for every Claude / ChatGPT / Gemini interaction: at maturity, millions of attestations per day, smoothly distributed, each attestation worth fractions of a cent in fee. **Sovereign-identity proofs** are government-issued identity assertions: hundreds per day, sparse, each attestation potentially gating access to financial services, government benefits, or compliance reporting and therefore worth dollars in fee. The two schemas share a chain but share nothing about their demand curves: volume differs by four orders of magnitude, fee elasticity differs by three, time-of-day distribution differs in shape entirely.

A single base fee cannot serve both. If the chain prices for the high-volume schema, the identity schema either underpays (and its bandwidth is squeezed by transient surges in the AI-provenance workload) or overpays (and the chain extracts more from identity users than the workload justifies). If the chain prices for the identity schema, the high-volume schema's users see fee swings driven by other workloads they have nothing to do with. The misalignment cost compounds at scale.

The thesis of this paper: the **schema is the natural fee-market unit on attestation-native chains**. The chain already accounts per-schema for attestation work (PoUA §4.3 reputation update is schema-tagged, schema-bound tokens depend on per-schema attestor sets, the schema registration mechanism creates first-class chain state per schema). Per-schema fee state is a natural extension of state that already exists, and per-schema isolation means a congested schema's base fee no longer drags an uncongested schema's users.

1.2 Why Now

Three independent developments converge to make per-schema fee markets implementable and desirable in 2026.

EIP-1559 is mature. Ethereum mainnet has shipped EIP-1559 since August 2021. The base-fee adjustment formula $b(t+1) = b(t) \cdot (1 + \xi \cdot (u - T)/T)$ with $\xi = 1/8$ has been live in production for five years. Convergence behavior, attack-surface, and parameter calibration are well-understood. We can borrow the dynamics and apply them per-schema with high confidence in stability.

Multi-resource fee markets are an active research area. Solana's priority fees, Cosmos's `x/feemarket` module, Aptos's gas pricing, and the EIP-7732 / EIP-4844 multi-dimensional fee proposals on Ethereum have each explored varieties of multi-resource pricing. The conceptual space is mapped; what is missing is a chain that treats workload type (rather than compute / storage / DA) as the resource being priced. That is what per-schema fees does.

Attestation chains have per-schema state at the consensus layer. PoUA's reputation accounting is schema-tagged. Schema-bound tokens use per-schema attestor sets as mint authority. Native delegation grants are schema-scoped (per the companion paper §3.3). Adding fee-market state per-schema is the natural completion of a state model that is already per-schema everywhere else.

Iris is the bellwether product. Iris's USD-billed-relayer model needs predictable per-schema fee dynamics: an Iris customer running a Themisra session needs to know what the per-attestation cost will be over the next 24 hours, not what the chain-wide average will be. A unified fee market makes Iris's subscription pricing harder; a per-schema fee market lets Iris hedge per-schema. The product-economic case for per-schema fees is concrete and immediate.

1.3 The Misalignment Problem

A unified base fee, like EIP-1559's, adjusts based on chain-wide utilization. The mechanism's elegance comes from a single signal: when the chain is congested, prices climb; when it is empty, prices drop. The economic intuition is clear.

The same mechanism, applied to a chain with heterogeneous workloads, produces three failure modes.

Failure mode 1: high-volume schema absorbs spikes in low-volume schema. Suppose the chain runs a steady Themisra workload at 80% utilization (well above the 50% target). The base fee climbs to extract revenue and discourage congestion. Now an identity-proof spike happens (a wave of new-user onboarding flows after a regulatory event). The spike adds 5% to chain-wide utilization, pushing it to 85%. Under a unified fee market, the base fee climbs further. Themisra users, who had nothing to do with the identity spike, pay more because of unrelated activity.

Failure mode 2: low-volume schema cannot trigger its own price discovery. Same chain, reverse problem. The identity workload spikes briefly (a few hundred attestations in a short window), but the chain-wide utilization barely moves because Themisra dominates the denominator. The identity spike never gets the price signal it needs to clear quickly. Identity users wait. The high-value low-volume schema does not get bandwidth proportional to its willingness to pay.

Failure mode 3: aggregate base fee fits no profile. The chain's steady-state base fee, optimized for an average across schemas, is too high for the high-volume low-margin schema and too low for the low-volume high-value schema. Both workloads pay the wrong price all the time, with the wrong price defined as the price the workload would have paid under a fee market sized to its own demand profile.

All three failure modes share a structural cause: the chain has more information about workload heterogeneity than the unified fee market can express. Per-schema state already exists in the chain; the fee market should use it.

1.4 The Central Question

> **How can an attestation-native chain price attestation work in a way that respects per-schema demand heterogeneity, while preserving PoUA's cost-to-grind security floor and composing cleanly with sponsored-gas and native-delegation patterns?**

This paper answers: per-schema EIP-1559 with a configurable burn-versus-routing split, coupled to PoUA's adaptive τ_{burn} rebase so that the chain-wide cost-to-grind floor is preserved even under per-schema isolation.

1.5 Approach in Brief

Each registered schema carries its own fee-market state: base fee b_σ, current utilization u_σ, target utilization T_σ, plus a small parameter vector for adjustment rate, tip floor, base-fee clip bounds, and a routing fraction ρ_σ. Per-block, the chain adjusts each schema's base fee independently using the EIP-1559 formula, scoped to that schema's observed utilization. Tips compete within schema capacity; cross-schema isolation is preserved by allocating per-block attestation slots per schema (proposer chooses the per-block allocation under governance-tuned bounds).

The burn split is the bridge to PoUA. Of every paid base fee, fraction τ_{burn} is burned (the same chain-wide burn that PoUA §4.4 governs), fraction $(1-\tau_{\text{burn}})\cdot\rho_\sigma$ flows to the schema registrant, and fraction $(1-\tau_{\text{burn}})\cdot(1-\rho_\sigma)$ flows to the validator who included the attestation. Tips flow entirely to the validator. Section 5 shows that even at the most aggressive routing ($\rho_\sigma = 0.5$ chain-wide), PoUA's Lemma 1 cost-to-grind floor is preserved with the same constants.

The mechanism composes orthogonally with two other primitives. **Sponsored gas** lets a third party pay base fee and tip on behalf of an attestor (Iris's pattern): signer and fee-payer are independent transaction fields, and the §4.3 sponsored-gas accounting handles the variance hedging. **Native delegation** (companion paper) lets a master key authorize a hot key to sign attestations within bounded scope; the per-schema fee market does not change anything about who can sign, only about who pays and how the chain prices the work.

1.6 Contributions

The paper makes four contributions.

A **mechanism specification** in §4: the per-schema base-fee adjustment formula, the per-block target-utilization calibration table, the tip auction within schema capacity, the burn-and-routing split, and the integration points with PoUA reputation accounting. The mechanism is fully specified at a level where a chain implementer can build it without further ambiguity.

A **cost-to-grind preservation theorem** in §5: under the per-schema fee market with arbitrary schema-set Σ and routing fractions $\{\rho_\sigma\}_{\sigma\in\Sigma}$ bounded to $\rho_\sigma \leq 0.5$, PoUA's §5.5.3 Lemma 1 holds with the same constants. The chain's economic-security floor is unchanged by introducing per-schema fees, even at the most adversary-friendly routing setting.

A **comparative analysis** in §6 positioning per-schema fees against EIP-1559, Solana priority fees, Cosmos `x/feemarket`, and Aptos gas pricing across five axes (granularity of pricing, target-utilization mechanism, burn fraction, sponsor-friendly extensions, multi-workload accommodation). Per-schema fees occupies a distinct point in this design space: the only mechanism in the comparison that treats workload type as the unit of pricing.

A **calibration recommendation** in §4.2 (taxonomy of three demand profiles plus latency-critical addendum) and §7.2 (v0 protocol parameters). The recommendation is governance guidance, not a hard constraint; the chain enforces a $[0.1, 0.9]$ bound on T_σ and a $[0, 0.5]$ bound on ρ_σ, but schemas declare their own values at registration within those bounds.

1.6.1 Status of Claims Following PoUA v0.7's discipline of separating claim categories explicitly:

Proven (formal mathematical argument under standard cryptographic and BFT assumptions):

- The §5.1 cost-to-grind preservation theorem proves PoUA's Lemma 1 floor holds under per-schema fees with $\rho_\sigma \leq 0.5$, with the same constants and the same security argument.

Empirical or heuristic, requiring devnet validation:

- The §4.2 calibration table (recommended T_σ per demand profile) is an architectural default. Real-world demand curves for live schemas will refine these targets once devnet operation produces data.
- The §5.4 sponsored-gas adversarial model assumes a specific Iris pricing pattern (pre-committed subscription with retroactive reimbursement above threshold). Other sponsorship models exist and need their own analysis.

Bounded under stated assumptions, where the assumptions are non-trivial and named:

- The §5.2 cross-schema censorship bound assumes the PoUA §A.1 KL-divergence detector's tolerance is calibrated against the chain-wide null distribution of schema arrivals. If the calibration is too loose, cross-schema censorship goes undetected; if too tight, honest fee-driven preference is flagged. Default tolerance from PoUA v0.7 §A.4 holds; per-schema fees do not require recalibration.
- The §3.2 validator income decomposition assumes the proposer is the validator who includes the attestation; in a separated-proposer-and-builder architecture (e.g., MEV-Boost-style), the income flows differently. The mechanism still works but §6.2 needs a separate treatment.

1.7 Scope and Non-Goals

In scope:

- Per-schema fee dynamics: base-fee adjustment, target utilization, tip auction, burn split, fee routing
- Integration with PoUA reputation accounting and adaptive τ_{burn} rebase
- Cost-to-grind preservation under per-schema isolation
- Composition with sponsored gas (Iris paymaster pattern)
- Composition with native delegation (companion paper)
- Cross-schema attack analysis and detector calibration

Explicitly out of scope:

- **Multi-resource fee markets.** A separate paper would price gas / storage / DA bandwidth as distinct resources. This paper prices workload-type (the schema) as a single composite resource and treats other resources as derived.
- **MEV (maximal extractable value).** Block-builder ordering preferences, sandwich attacks, and reorg incentives are outside this paper. The §6.2 builder-side incentive analysis touches MEV briefly but defers detailed treatment.
- **Cross-chain fee-market portability.** A schema registered on Ligate Chain that is also recognized on a counterparty chain via IBC needs cross-chain price discovery. Out of scope for v0.2; a follow-on paper will cover it.
- **EVM-execution fee market.** If Ligate Chain's v4 phase ships EVM execution (per the chain roadmap, deliberately deferred), EVM contract calls will have their own fee dynamics that this paper does not address.

1.8 Document Structure

Section 1.6.1 separates the paper's claims into proven, bounded-under-stated-assumptions, and empirical-or-heuristic; readers in a hurry may want to start there. Section 2 surveys EIP-1559, multi-resource fee market designs, and attestation-chain fee patterns as background. Section 3 fixes the system model: the schema as fee-market unit, validator income decomposition, demand profile taxonomy. Section 4 specifies the mechanism. Section 5 proves the cost-to-grind preservation theorem and analyzes four threat models. Section 6 analyzes incentives across three parties (validator, builder, sponsor). Section 7 walks through the Sovereign SDK integration and recommended v0 parameters. Section 8 positions per-schema fees against prior systems. Section 9 lists limitations and future work; Section 10 concludes; Section 11 collects frequently asked questions.

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2. Background and Related Work

Fee market design has gone through three distinct generations on production chains: per-transaction auction (Bitcoin and pre-1559 Ethereum), unified base fee with target utilization (EIP-1559), and multi-resource fee markets (Solana, Aptos, Cosmos `x/feemarket`). This section surveys each and then positions attestation-chain fee patterns (EAS, Ceramic, EigenLayer AVS) against the per-schema thesis.

2.1 EIP-1559

Ethereum Improvement Proposal 1559 (Buterin et al., 2021) replaced Ethereum's first-price gas auction with a base-fee + tip mechanism. Each block carries a base fee determined by the previous block's utilization, with target utilization $T = 0.5$ (half-full target). The per-block adjustment rule is:

$$b(t+1) = b(t) \cdot \left(1 + \xi \cdot \frac{u(t) - T}{T}\right), \quad \xi = \frac{1}{8}$$

allowing a maximum $\pm 12.5\%$ swing per block. The base fee is burned (sent to a verifiably unspendable address); tips go to the proposer. The mechanism ships convergence properties around $u = T$ that have held in production since August 2021.

What EIP-1559 does well. Base-fee burn anchors the chain's monetary policy: high demand burns more, deflationary pressure aligns with usage. Target utilization gives predictable headroom (50% buffer over steady state, supporting normal-load latency guarantees). Per-block adjustment is bounded ($\xi = 1/8$ prevents oscillation), and the mechanism is composable with sponsored-gas patterns at the application layer (ERC-4337 paymasters).

What EIP-1559 cannot do that per-schema fees can. A single base fee assumes a single workload. Ethereum's heterogeneity (token transfers, DeFi, DA blobs since EIP-4844) has driven the chain toward separate fee markets for DA (blobspace) while keeping execution as one fee market. The trend is per-resource separation; this paper extends it to per-workload (schema) separation.

2.2 Multi-Resource Fee Markets

Three production chains have implemented variants of multi-dimensional pricing.

Solana charges a per-compute-unit (CU) fee with optional priority fees. The base unit price is set per-transaction by the submitter, subject to chain-enforced minimum and the proposer-tunable maximum CU per block. Priority fees compete for inclusion within the block. The mechanism is closer to a first-price auction than to EIP-1559's adjusting base fee; convergence happens via signaling between users and proposers rather than via a deterministic curve.

Cosmos `x/feemarket` (cosmos-sdk v0.50+) provides an EIP-1559-inspired adjusting base fee at the chain level, with optional per-module fee multipliers. Each module (the `bank` module, the `staking` module, custom application modules) can declare a fee multiplier that scales the chain-wide base fee for transactions of that module type. This is the closest existing primitive to per-schema fees: per-module multipliers approximate per-workload pricing. The differences from this paper: (i) Cosmos multipliers are governance-set static parameters, not dynamic per-module utilization-driven, (ii) Cosmos has no per-module target-utilization mechanism, (iii) the burn-and-routing split is global, not per-module.

Aptos gas pricing uses a per-transaction gas-unit-price plus a max-gas-amount. The proposer admits transactions in tip-priority order. The mechanism is closer to a sealed-bid auction than to a smoothly-adjusting market. Aptos has no per-module differentiation; one pool of gas, one curve.

EIP-4844 blob fees on Ethereum. Independent of execution gas, blob transactions (data-availability blobs for L2s) have their own EIP-1559-style fee market. The blob base fee adjusts based on blob utilization, completely separate from execution-gas utilization. This is the first production example of *per-resource fee market separation* on a major chain. Per-schema fees extends the same idea from per-resource to per-workload granularity.

What this paper takes from prior work. The EIP-1559 adjustment formula. The Cosmos per-module multiplier intuition (per-module pricing matters). The EIP-4844 lesson that per-resource fee separation works in production. The Aptos / Solana lesson that simpler designs (per-transaction auctions) suffer at the heterogeneity edge.

What this paper does that none do. Make the schema (the workload type) the unit of fee adjustment, with full per-schema state (not just per-module multiplier), full per-schema target utilization, and a burn-versus-routing split that flows part of the non-burned base fee to the schema registrant.

2.3 Attestation Chains and Schema Heterogeneity

Three attestation-adjacent systems are worth comparing on the schema-heterogeneity axis.

Ethereum Attestation Service (EAS) is a smart-contract attestation framework on Ethereum. Schemas are first-class on-chain entities, but EAS inherits Ethereum's unified EIP-1559 fee market: a single base fee plus tip applies to all EAS transactions regardless of schema. This is exactly the misalignment §1.3 describes. EAS's design is reasonable for Ethereum (workloads are dominated by DeFi, not attestations), but a chain whose primary workload *is* attestations cannot afford to mix the fee surface.

Ceramic uses per-stream rentals (model fees) rather than per-attestation fees. Each Ceramic Model has a fee curve tied to the model's storage footprint and update frequency. This is closer to a SaaS pricing model than a public-chain fee market: the per-model price is set by the model author and doesn't adjust to demand. The intuition (per-workload pricing matters) is right; the mechanism (static fee curves) doesn't fit chains that need dynamic price discovery.

EigenLayer AVS economics. Restaking AVSs (Actively Validated Services) have per-AVS slashing and reward economics: an AVS author specifies the rewards / slashing parameters for operators serving that AVS. This is a different axis (security economics, not transaction fees), but the analogy is structural: per-AVS economics is to restaking as per-schema fees is to attestation chains. Both treat the workload as the natural unit of economic differentiation.

Ligate's position. Per-schema fees as a runtime primitive, with full per-schema dynamic adjustment (not static like Ceramic, not global like EAS, not security-economics-flavored like EigenLayer). The schema is the workload, and the workload is the fee-market unit. The §6 comparison table puts this paper alongside EIP-1559, Cosmos `x/feemarket`, Solana priority fees, and Aptos gas pricing across six axes; per-schema fees occupies a distinct design point.

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3. System Model

3.1 The Schema as Fee-Market Unit

A schema σ is a registered, named template on Ligate Chain. The PoUA paper (v0.8 §3.4) defines a schema as a tuple $(\text{schema\_id}, \text{attestor\_set}, \text{validation\_rule}, \text{fee\_routing\_bps}, \text{metadata})$ where `schema_id` is the canonical hash, `attestor_set` is the threshold-signature group authorized to attest under it, and `validation_rule` is the validity predicate. Each registered schema persists in chain state as a first-class entity. We extend the schema record with a **fee-market state**:

$$\text{FeeState}(\sigma) = (b_\sigma, u_\sigma(t), T_\sigma, \rho_\sigma, \tau_\sigma^{\min}, b_\sigma^{\min}, b_\sigma^{\max}, \xi_\sigma)$$

where:

- $b_\sigma \in [b_\sigma^{\min}, b_\sigma^{\max}]$ is the current per-schema base fee, denominated in the chain's micro-unit (e.g., `uavow` once the rename in ligate-chain#457 lands).
- $u_\sigma(t) \in [0, 1]$ is the schema's observed utilization at block t: the fraction of σ's allocated attestation slots filled in that block.
- $T_\sigma \in [0.1, 0.9]$ is the target utilization (§4.2 calibration table).
- $\rho_\sigma \in [0, 0.5]$ is the fee-routing fraction: the share of non-burned base fee that flows to the schema registrant. Default $\rho_\sigma = 0$.
- $\tau_\sigma^{\min} \geq 0$ is the tip floor: the minimum tip an attestation must offer to be admitted (§4.3).
- $b_\sigma^{\min}, b_\sigma^{\max}$ are governance-tunable clip bounds on the base fee.
- $\xi_\sigma \in (0, 1]$ is the per-block max-change rate (default $\xi_\sigma = 1/8$, matching EIP-1559).

Why the schema is the natural unit. Two arguments. First, **demand profiles differ by orders of magnitude per-schema** (§3.3). A single chain-wide base fee penalizes the low-demand schema with the high-demand schema's congestion, and vice versa. The fee market should isolate demand signals at the same granularity that the chain isolates attestation work: per-schema.

Second, **the chain already maintains per-schema state for PoUA reputation accounting** (PoUA §4.3: g_v accumulates from valid attestation work, schema-tagged). Adding fee-market state per-schema is a natural extension of state that already exists. No new global-state surface; no new module boundary; the schema record gets four additional fields.

State-tree cost. FeeState(σ) adds approximately 64 bytes per schema (one `u64` for b_σ, one for u_σ as a fixed-point fraction, eight u32 / u16 fields for the rest). At 1,000 registered schemas, this is $\sim$ 64 KB of state per epoch. Negligible against the chain's expected state size at devnet + post-devnet scale.

Registration cost. Per PoUA §4.4, registering a new schema costs `RegisterSchema` fee plus the configured `RegisterAttestorSet` fee. The fee-market state is initialized at registration with $(b_\sigma, u_\sigma, T_\sigma) = (b_\sigma^{\min}, 0, T_{\text{default}})$ and the registrant's declared parameters for $(\rho_\sigma, \tau_\sigma^{\min}, b_\sigma^{\min}, b_\sigma^{\max}, \xi_\sigma)$. No additional registration cost is added by this paper; the existing `RegisterSchema` fee absorbs the new state.

3.2 Validator Income Decomposition

A validator v proposing block B at slot t earns income from three streams. Let $|B| = \{\alpha : \alpha \in B\}$ be the set of attestations included in B. Then:

$$R_v(B,t) = R_b + \sum_{\alpha\in B} \tau_\alpha + \sum_{\alpha\in B} (1-\tau_{\text{burn}}) \cdot (1-\rho_{\sigma(\alpha)}) \cdot b_{\sigma(\alpha)} \cdot |\alpha|$$

where:

- R_b is the **protocol block reward** (chain-wide constant, set by governance; in PoUA v0 set as a small per-block emission until R_f stabilizes).
- τ_α is the **tip** for attestation α, flowing entirely to the proposer.
- $\sigma(\alpha)$ is the schema of attestation α.
- $b_{\sigma(\alpha)}$ is the current per-schema base fee at slot t for α's schema.
- $|\alpha|$ is the size of the attestation in attestation-units (typically 1; see §3.3 for multi-unit attestations).
- τ_{burn} is the chain-wide burn fraction governed by PoUA §4.4.2's adaptive rebase. The complement $(1-\tau_{\text{burn}})$ is the non-burned share.
- $\rho_{\sigma(\alpha)}$ is the schema's fee-routing fraction. The complement $(1-\rho_{\sigma(\alpha)})$ is the validator's share of the non-burned amount.

The third term is the **validator's per-attestation base-fee share**, which depends on which schemas the validator includes. A validator who proposes blocks heavy in high-fee schemas earns more from this term; a validator who specializes in low-fee schemas earns less from this term but accumulates reputation just as fast (PoUA reputation update is fee-weighted by total fee, not validator-share-of-fee).

Three rebate destinations per paid base fee. Decomposing the base fee for one attestation α of schema σ:

Destination	Fraction of b_σ	Recipient
Burn	τ_{burn}	Pure-burn destination (PoUA §5.5.3)
Schema registrant (routing)	$(1-\tau_{\text{burn}}) \cdot \rho_\sigma$	Schema's declared `fee_routing` address
Validator (proposer)	$(1-\tau_{\text{burn}}) \cdot (1-\rho_\sigma)$	Block proposer

Tips τ_α are separate and flow 100% to the proposer.

Why the validator's income is fee-schema-mixed. A validator's per-block revenue depends on the schema mix in the block. This creates an incentive for the validator to prefer high-fee-schema attestations during proposal. PoUA §A.1's KL-divergence detector bounds this preference: the validator's schema-mix distribution is detected against the chain-wide null. A validator who deviates from the null distribution (e.g., consistently underweighting low-fee schemas) is flagged. The detector is the protocol-level enforcement that **fee-driven inclusion preferences cannot turn into censorship of low-fee schemas**.

§5.2 and §6.1 take this further: we show that under standard parameters, the §A.1 detector tolerance is wide enough to permit honest fee-driven preference but tight enough to flag adversarial cross-schema censorship.

3.3 Demand Profile Taxonomy

Attestation chains do not have one demand profile; they have several, and the per-schema design is justified by how different they are. We taxonomize three profiles and assign each a recommended target utilization.

Profile A: High-volume, low-variance. Steady-state attestation workloads with millions of attestations per day, smoothly distributed across time. Examples:

- **Themisra `proof-of-prompt/v1`** at maturity: every Claude / ChatGPT / Gemini session ends with an attestation submission. Demand correlates to user activity, smooth on multi-hour scales.
- **Content-authenticity attestations** for media pipelines: every uploaded video / image generates one attestation per processing step. Continuous load.

These profiles have low per-block variance and benefit from a base fee that prices steadily at the demand curve. Recommended $T_\sigma = 0.5$ (default): balances predictability against revenue extraction. The 2x headroom over steady-state demand absorbs normal noise without triggering rapid base-fee swings.

Profile B: Low-volume, high-value. Rare attestations with high per-claim economic significance. Examples:

- **Sovereign-identity proofs**: government-issued identity assertions, attestor set is a small set of accredited identity providers. Demand is correlated to onboarding flows, hundreds per day at maturity.
- **Regulatory filings**: per-quarter or per-event compliance attestations. Sparse, predictable.

These profiles want to pack closer to demand peak (high T_σ) because demand is predictable and the cost of paying slightly more for fast inclusion is low against the per-attestation value. Recommended $T_\sigma = 0.7$: 1.4x headroom over steady-state demand, with the higher utilization extracting revenue closer to the demand curve.

Profile C: Bursty / event-driven. Demand spikes 10x to 100x baseline for short windows around events. Examples:

- **NFT mint attestations** around drops: a single project may generate tens of thousands of attestations in a five-minute window, then return to baseline.
- **News-event-driven content provenance**: breaking news / live events / political moments generate concentrated bursts of `themisra.content-provenance/v1` attestations.
- **Iris-agent campaigns**: an Iris-hosted agent running a coordinated multi-hour campaign generates a burst of attestations across multiple schemas.

These profiles need heavy headroom. A spike that triggers base-fee climbing punishes legitimate users who happen to participate in the spike. Recommended $T_\sigma = 0.3$: 3.3x headroom over steady-state demand, prioritizing predictability for users during the burst over revenue extraction.

Latency-critical addendum. Schemas with hard latency requirements (e.g., Iris agent actions where the user is waiting on a sub-second response) want $T_\sigma = 0.3$ even at low expected volume, to maintain immediate-inclusion guarantees during normal load. Latency is treated as a special case of bursty (the "burst" is occasional unexpected load).

The taxonomy in one table:

Profile	Example	Recommended T_σ	Recommended ρ_σ
A: High-volume, low-variance	Themisra prompts	0.5	0.1-0.3
B: Low-volume, high-value	Sovereign identity	0.7	0.3-0.5
C: Bursty / event-driven	NFT mints, Iris campaigns	0.3	0.1-0.3
Latency-critical (special case)	Iris agent actions	0.3	0-0.1

These are architectural defaults. Devnet observation in v0.2.x will refine per-schema targets; schemas declare a preferred T_σ at registration; the chain enforces the protocol-level $[0.1, 0.9]$ bound. The recommendation table is governance guidance, not a hard constraint.

Attestation-unit granularity. Within a schema, attestations may carry weight $|\alpha| > 1$ for compound claims (a single attestation that references k sub-claims). The fee mechanism treats $|\alpha|$ as a multiplier on base fee and tip: an attestation of size $|\alpha| = 3$ pays $3 \cdot b_\sigma + \tau_\alpha$ (the tip is per-attestation, not per-unit). This matches the rest of the chain's resource-accounting convention.

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4. Per-Schema Fee Mechanism

This section specifies the per-schema fee market: how the base fee adjusts in response to schema-specific demand, how priority within a schema is auctioned via tips, how base-fee revenue is split between burn and validator reward, and how the mechanism integrates with PoUA reputation accounting.

The mechanism is a per-schema instantiation of EIP-1559's adjustment dynamics, with three modifications appropriate to attestation-native chains: (1) per-schema state rather than global state, (2) integration with PoUA's adaptive τ_{burn} rebase from PoUA §4.4.2, and (3) explicit accounting for sponsored-gas relayers (used by the Iris MCP relayer, design-partner-validated).

4.1 Base-Fee Adjustment Formula

For each registered schema $\sigma \in \Sigma$, the chain maintains a per-schema base fee $b_\sigma(t)$, target utilization T_σ, and observed utilization $u_\sigma(t)$. Per-block adjustment follows the EIP-1559 form:

$$b_\sigma(t+1) = b_\sigma(t) \cdot \left(1 + \xi \cdot \frac{u_\sigma(t) - T_\sigma}{T_\sigma}\right)$$

clipped to $[b_\sigma^{\min}, b_\sigma^{\max}]$, where:

- $u_\sigma(t) \in [0, 1]$ is the fraction of schema σ's allocated attestation slots filled in block t
- $T_\sigma \in (0, 1)$ is the schema's target utilization (§4.2)
- ξ is the per-block max-change parameter
- $[b_\sigma^{\min}, b_\sigma^{\max}]$ are governance-tunable clip bounds

Recommended ξ. $\xi = 1/8$ matches EIP-1559's calibration (max $\pm 12.5\%$ per block). At the v0 block time of 12 seconds, this gives a $\sim 64\%$ swing potential per epoch (6 blocks of 1.125); aggressive enough to track demand shifts, gentle enough to avoid oscillation.

Allocated slots per schema. A block's total attestation capacity is partitioned across schemas at proposer-selection time. The simplest allocation is fixed-share-per-schema (each schema gets $\lfloor C_{\text{block}} \cdot w_\sigma \rfloor$ slots, $\sum w_\sigma = 1$); a richer allocation is governance-tunable per-schema utilization-aware. v0 ships fixed-share as the default.

Convergence and stability. The dynamics admit a fixed point at $u_\sigma^* = T_\sigma$. Around this fixed point, perturbations decay geometrically with rate $1 - \xi$ at first order. The mechanism inherits EIP-1559's convergence behavior; per-schema isolation means a high-utilization schema cannot drag down the base fee of a low-utilization schema. This is the central architectural advantage of the per-schema design.

Drift interaction with PoUA τ_{burn} rebase. Both this paper's per-schema base fee and PoUA §4.4.2's τ_{burn} are time-varying parameters. The composition is hierarchical: τ_{burn} governs what fraction of b_σ is burned (§4.4); b_σ itself adjusts per-schema. The v0.8 PoUA paper §4.4.3 spec at `papers/poua/specs/eta-lambda-rebase.md` §5.2 verifies that the two rebases are first-order independent: schema-fee drift is the input to b_σ, while cost-to-grind drift is the input to τ_{burn}.

4.2 Target Utilization Calibration

The target utilization T_σ controls the trade-off between **predictability** (low T_σ, more headroom for spikes) and **revenue extraction** (high T_σ, prices closer to demand peak).

Recommended default: $T_\sigma = 0.5$. Matches EIP-1559's choice. Provides $2\times$ headroom over steady-state demand.

Schema-specific deviations.

Schema profile	Recommended T_σ	Rationale
Stable, low-volume (e.g., regulatory filings)	0.7	Variance is low; can pack closer to demand peak without congestion
High-volume, low-variance (e.g., Themisra prompts)	0.5	Default; balances predictability and revenue
Bursty (e.g., Kleidon mint events around drops)	0.3	Heavy headroom to absorb spikes without triggering rapid base-fee climbs
Real-time / time-critical (e.g., Iris agent actions)	0.3	Latency sensitivity demands immediate-inclusion guarantees during normal load

[**Measured at v0.2.5+:** these are architectural defaults; devnet observation will refine per-schema targets. Schemas can declare a preferred T_σ at registration; the chain enforces governance bounds.]

Bounds. T_σ is bounded to $[0.1, 0.9]$ at the protocol level. A schema requesting $T_\sigma < 0.1$ is asking for $> 10\times$ headroom over peak demand, which is wasteful; a schema requesting $T_\sigma > 0.9$ leaves no buffer for normal variance and would oscillate. Governance can adjust the bound but not the rate-limit (max one bound change per quarter).

4.3 Tip Mechanism

Within a schema's allocated capacity, attestations are admitted in order of tip-per-attestation τ_α. Tips go to the proposing validator and are not subject to the base-fee burn (§4.4). The mechanism is identical in structure to EIP-1559's priority fee; per-schema partitioning means tips compete only against same-schema attestations.

Tip floor. Schemas may declare a minimum tip $\tau_\sigma^{\min}$ at registration to prevent zero-fee spam. Default $\tau_\sigma^{\min} = 0$ (open to economic-signaling-only).

Tip auction within block. The proposer admits attestations greedily by descending τ_α until the schema's allocated capacity is filled. Ties broken by first-seen mempool ordering. The greedy auction is welfare-suboptimal compared to a sealed-bid second-price design but matches EIP-1559's actual deployment behavior; v2 extensions can revisit.

Sponsored gas (paymaster pattern). Iris-style relayers pay tips on behalf of agents that don't hold `$AVOW`. The protocol semantics are unchanged: a single signed transaction can declare a *fee payer* address distinct from the *attestor* address. The fee payer pays both base fee and tip; the attestor signs the attestation content. This composes orthogonally with native delegation (companion paper, v0.2): a delegated hot key can submit attestations whose fees are paid by a third-party paymaster. v2 of this paper details the formal fee-payer mechanism; v0.2 establishes that the design admits it cleanly.

4.4 Base-Fee Burn

Of every paid base fee $b_\sigma \cdot |\alpha|$ for an admitted attestation α of schema σ:

- A fraction τ_{burn} is burned (sent to the pure-burn destination per PoUA §5.5.3)
- A fraction $1 - \tau_{\text{burn}}$ is rebated to the schema's fee-routing address (the registrant; up to 50% per `OVERVIEW.md`'s schema primitive)
- Tips τ_α are unaffected; they go to the proposing validator

Why the schema gets the non-burned fraction. The schema registrant pays the registration fee at schema-creation, including the cost of operating their attestor set. Routing a fraction of base fees to the registrant aligns incentive: schemas that drive volume earn proportional ongoing revenue, which funds attestor-set operations and incentivizes schema curation. This is the "build a chain on top of Ligate" path: a partner registers a schema, pays the registration fee, and earns ongoing fee revenue from that schema's traffic.

Per-schema fee-routing parameter. At registration, a schema declares its fee-routing fraction $\rho_\sigma \in [0, 0.5]$ (capped at 50% per protocol parameter). Effective burn is then:

$$\text{burn}(\sigma) = \tau_{\text{burn}} + (1 - \tau_{\text{burn}}) \cdot (1 - \rho_\sigma)$$

with the remainder $(1 - \tau_{\text{burn}}) \cdot \rho_\sigma$ routed to the schema registrant.

Interaction with PoUA τ_{burn} rebase. τ_{burn} is the chain-wide burn fraction governed by PoUA §4.4.2's adaptive rebase. The per-schema ρ_σ is set at registration and changes only via governance proposal. The two parameters are independent: τ_{burn} tracks economic-security drift, ρ_σ tracks schema-builder economics.

Lemma 1 cost-to-grind preservation. PoUA §5.5.3's Lemma 1 bounds adversary cost-to-grind as $F_{\text{net}} \geq \tau_{\text{burn}} \cdot \Delta r/(\eta \cdot \alpha_{\text{eff}})$. The per-schema fee market does not weaken this bound: even if every schema sets $\rho_\sigma = 0.5$, the burned fraction per attestation is at least $\tau_{\text{burn}} \cdot 0.5 \cdot b_\sigma$, and the cost-to-grind floor scales with that burned amount. The paper's central security claim survives the introduction of per-schema fee routing.

4.5 Integration with PoUA

The per-schema fee market interacts with PoUA reputation in three places.

Reputation accumulation is schema-agnostic. Validators accumulate reputation g_v from valid attestation work across all schemas, weighted by fee paid (PoUA §4.3):

$$G_v^{\text{prop}}(t) = \sum_{B \in \text{Proposed}_v(t,t+E)} \sum_{\alpha \in B} \mathbb{1}[\alpha \text{ valid}] \cdot \text{fee}(\alpha)$$

where $\text{fee}(\alpha) = b_\sigma + \tau_\alpha$ is the total fee paid (base + tip) for attestation α of schema σ. A validator who proposes blocks heavy in high-fee schema attestations accumulates reputation faster than one who specializes in low-fee schemas, but both accumulate strictly more than zero. This preserves the §6.2 honest-equilibrium incentive structure.

Validator income decomposition. Per PoUA §6.1, validator income from block B at slot t is:

$$R_v(B,t) = R_b + \sum_{\alpha \in B} \tau_\alpha + \sum_{\alpha \in B} (1 - \tau_{\text{burn}}) \cdot (1 - \rho_{\sigma(\alpha)}) \cdot b_{\sigma(\alpha)}$$

where R_b is the protocol block reward, τ_α is the per-attestation tip, and the third term is the validator's share of the per-schema base fee after burn and schema-routing. The validator's revenue depends on schema mix, but the §A.1 KL-divergence detector enforces that schema mix tracks the chain-wide null distribution; a validator preferentially including high-fee-schema attestations gets flagged.

Detector calibration accommodates per-schema fees. PoUA §A.1 defines a KL-divergence detector against a chain-wide schema distribution null. With per-schema fees, validator-utility incentivizes schema-mix deviation; the detector calibration (see the `detectors.py` module under `prototypes/poua-sim`) tracks the actual per-schema arrival distribution. v0.2 of this paper specifies the joint calibration: detector null = empirical per-schema arrival distribution at the chain level, not at the validator level. v0.7.2 PoUA §A.1 already supports this; the per-schema fees mechanism does not require detector revision.

Sponsored gas does not inflate reputation. Iris-style relayers paying base fees and tips on behalf of agents do not accumulate reputation themselves (they are not validators). The agent submitting the attestation gets credited (reputation accumulates against the *attestor*, not the *fee payer*); this matches the paper's intent that reputation tracks productive work, not financial sponsorship.

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5. Security Analysis

We analyze the per-schema fee market against five threat models. The first (§5.1) is the central theorem: PoUA's cost-to-grind floor is preserved under per-schema isolation. The remaining four (§5.2-§5.5) analyze concrete cross-schema attack patterns and their defenses.

5.1 Cost-to-Grind Preservation Theorem

Claim. Under the per-schema fee market specified in §4, with arbitrary registered schema set Σ and per-schema routing fractions $\{\rho_\sigma\}_{\sigma \in \Sigma}$ bounded by $\rho_\sigma \in [0, 0.5]$, PoUA's §5.5.3 Lemma 1 cost-to-grind floor holds with the same constants:

$$F_{\text{net}}(\sigma) \geq \tau_{\text{burn}} \cdot \frac{\Delta r}{\eta \cdot \alpha_{\text{eff}}}$$

for every schema σ, where $F_{\text{net}}(\sigma)$ is the non-recoverable cost an adversary pays per unit of reputation gained through attestations of schema σ.

Setup. PoUA Lemma 1 bounds the cost an adversary must pay to inflate their reputation by Δr through legitimate-looking attestation work. The bound assumes a per-attestation fee floor that is partly burned (non-recoverable to the adversary) and partly paid out (potentially recoverable if the adversary's coalition includes both attestors and validators). The relevant constants:

- τ_{burn}: chain-wide burn fraction (PoUA §4.4.2, adaptively rebased)
- Δr: target reputation gain
- η: adversary's coalition share of validator power
- α_{eff}: effective reputation-per-fee ratio under coalition's attestation pattern

Lemma 1 establishes $F_{\text{net}} \geq \tau_{\text{burn}} \cdot \Delta r / (\eta \cdot \alpha_{\text{eff}})$ in the chain-wide setting where every paid base fee contributes τ_{burn} to burn and $(1 - \tau_{\text{burn}})$ to validator income.

The challenge of per-schema routing. Under per-schema fees, a fraction $(1 - \tau_{\text{burn}}) \cdot \rho_\sigma$ of every paid base fee flows to the schema registrant rather than to the validator. From an adversary's perspective, this fraction is "potentially recoverable" if the adversary controls the schema registrant address (which they would, if they registered the schema being attacked). The naive concern: routing reduces the non-recoverable share, weakening the cost-to-grind floor.

Proof of the bound. The burned fraction per paid base fee is exactly τ_{burn}, independent of ρ_σ. The routing fraction ρ_σ partitions the *non-burned* share between validator and schema registrant; it does not touch the burned share. Substituting into Lemma 1:

$$F_{\text{net}}(\sigma) = \tau_{\text{burn}} \cdot b_\sigma \cdot (\text{attestations submitted to gain } \Delta r)$$

The chain-wide formula $F_{\text{net}} \geq \tau_{\text{burn}} \cdot \Delta r / (\eta \cdot \alpha_{\text{eff}})$ holds per-schema with the *same* τ_{burn}. Per-schema base fees b_σ may differ from a hypothetical chain-wide base fee, but the proportionality between fee and reputation gained (α_{eff}) absorbs that difference. The adversary cannot reduce their cost-to-grind by registering a schema with their preferred ρ_σ: routing flows the *post-burn* amount, never the burned amount.

Corollary (worst case $\rho_\sigma = 0.5$ for every schema). Even if every registered schema sets $\rho_\sigma = 0.5$, half of the non-burned base fee flows to the schema registrant. The burned fraction is still τ_{burn} per attestation, the cost-to-grind floor is still $F_{\text{net}} \geq \tau_{\text{burn}} \cdot \Delta r/(\eta \cdot \alpha_{\text{eff}})$.

Discussion. The mechanism design choice that makes this work is keeping τ_{burn} chain-wide rather than per-schema. A per-schema burn fraction $\tau_{\text{burn},\sigma}$ would let the schema registrant trade burn for routing on their own schema, breaking the floor. The composition is: chain-wide τ_{burn} (PoUA §4.4.2 adaptive, tracking economic-security drift), per-schema ρ_σ (set at registration, tracking schema-builder economics). The two parameters live at different layers of the protocol and do not interfere.

5.2 Cross-Schema Arbitrage by Validator Inclusion Preference

Setup. A validator's per-block income (§3.2) depends on the schema mix in the block, because schemas have different b_σ and different ρ_σ. A profit-maximizing validator prefers to include attestations from schemas with the highest $(1-\tau_{\text{burn}})\cdot(1-\rho_\sigma)\cdot b_\sigma$ per unit of block capacity. Left unchecked, this preference could turn into censorship of low-fee schemas: a validator might consistently exclude low-fee attestations even when they offer competitive tips.

Defense (per-schema slot allocation). The mechanism (§4.1) allocates per-block attestation slots per schema, $\lfloor C_{\text{block}} \cdot w_\sigma \rfloor$ with $\sum w_\sigma = 1$. A validator cannot fill an entire block with high-fee attestations from one schema; they must respect the per-schema slot allocation. This caps the magnitude of inclusion preference at the slot-allocation granularity.

Defense (PoUA §A.1 KL-divergence detector). Within the per-schema slot allocation, a validator might still prefer high-fee schemas at the margin. PoUA's §A.1 detector tracks each validator's empirical schema-mix distribution against the chain-wide null. A validator who consistently underweights low-fee schemas (beyond what the slot allocation permits) is flagged. The default tolerance from PoUA v0.7 §A.4 was calibrated against a unified fee market; per-schema fees do not require recalibration because the null distribution is the chain-wide observed arrival distribution, which already incorporates per-schema fee dynamics.

Bound. A validator deviating from the null by more than δ in KL divergence triggers the §A.1 detector. From PoUA §A.4, $\delta = 0.5$ at the v0 calibration. This permits substantial honest fee-driven preference (a validator can favor high-fee schemas at 2-3x the null weight without triggering) while flagging coordinated cross-schema censorship.

Worst-case attack. An adversary controlling enough validator power could selectively exclude one schema entirely (drop its slot-allocated bandwidth). This is the classic single-schema censorship attack; PoUA §5.2 already covers it (safety inheritance plus the force-include path in ligate-chain). Per-schema fees do not introduce a new censorship surface; they inherit PoUA's existing defenses against single-schema censorship.

5.3 Base-Fee Manipulation by Validator Coalition

Setup. A validator coalition with control over a fraction η of proposer slots could selectively pump or suppress a target schema's base fee. Including high-tip attestations from schema σ artificially inflates u_σ, climbing b_σ; excluding attestations from σ artificially deflates u_σ, dropping b_σ. Either direction could be used to grief legitimate users of σ or to extract surplus value for the coalition.

Bound (max-change-per-block). The §4.1 adjustment formula clips per-block changes to $\pm\xi$ (default $\xi = 1/8$, max 12.5% swing). A coalition can move b_σ by at most ξ per block they propose.

Even with $\eta = 0.3$ (an adversary cartel), the expected swing per block is $0.3 \cdot \xi = 3.75\%$. Over an epoch of 6 blocks (12-second block time, 72-second epoch), the maximum cumulative swing is $\sim 25\%$ from the coalition's contribution. Bounded enough that legitimate users see a transient surge, not a permanent dislocation.

Bound (mempool transparency). b_σ adjustment is deterministic from observed utilization. An adversary cannot fabricate utilization out of thin air; they must actually include or exclude real attestations. Mempool observability means the manipulation is *visible* to honest validators: a sudden spike of high-tip attestations from one address pattern, included only by one validator cartel, is detectable by the §A.1 detector and by ad-hoc analysis of the mempool.

Bound (economic counterweight from tips). A coalition extracting surplus value through base-fee manipulation must absorb the cost of including high-tip attestations they themselves submit. The tip flows to *some* validator (the proposer); if the coalition is the proposer, they earn the tip back, but they also paid the base fee that gets burned. Net: the coalition pays τ_{burn} to burn for every attestation they use to manipulate the base fee, just like any other adversary running on a PoUA chain.

5.4 Fee-Griefing Across Schemas

Setup. An adversary submits high-tip attestations to a target schema σ_{target} to inflate $u_{\sigma_{\text{target}}}$ and drive up $b_{\sigma_{\text{target}}}$. Legitimate users of σ_{target} then face elevated fees. The attack cost is the adversary's tips (paid to proposers); the attack benefit is making the target schema expensive for legitimate users.

Defense (per-schema isolation). This attack pattern is contained within σ_{target}. Other schemas' base fees are unaffected. The blast radius of the attack is limited to the target schema's users, not the entire chain. Compared to the same attack pattern under a unified fee market (where griefing one workload climbs the chain-wide base fee), per-schema fees *reduce* the attack's blast radius.

Defense (max-change rate-limit). The same $\xi = 1/8$ bound applies. The adversary cannot move $b_{\sigma_{\text{target}}}$ faster than 12.5% per block. To sustain elevated $b_{\sigma_{\text{target}}}$, the adversary must keep submitting high-tip attestations indefinitely; the attack has linear cost in attack duration.

Defense (cost-to-grief vs cost-to-grieved). Each attack-attestation costs the adversary $b_\sigma + \tau_\alpha$. The legitimate user's price increase from one attack-attestation is small (one increment of $\xi/|B|$ in the next base fee). The cost ratio of attacker-spend to victim-cost-increase is on the order of $|B|/\xi$ per attestation, which under v0 parameters ($|B| \sim 100$, $\xi = 1/8$) is about 800. Each \$1 the attacker spends raises legitimate-user cost by \$0.0012 per attestation. The attack scales poorly.

Worst case. A well-resourced adversary running sustained griefing against a low-volume schema could double its base fee for hours. The legitimate users of that schema either pay the elevated fee, wait it out, or migrate to a different schema. The attack is unpleasant but bounded; the chain does not lose money (most of the attacker spend is burned), and the target schema's users have product-layer mitigations (longer time-bounds on grants, deferred submission, alternative schemas for the same use case).

5.5 Sponsored-Gas Adversarial Patterns

Setup. Iris pays base fees and tips on behalf of customer agents. The Iris pricing model is monthly USD subscription against expected per-attestation `$AVOW` cost over the billing period. An adversary running an Iris-customer agent could try to exhaust Iris's budget, or could cause Iris's pre-committed price curve to under-collect against actual costs.

Pattern A: budget exhaustion. Adversary agent submits attestation floods, consuming Iris's budgeted gas faster than the subscription allows. Iris must either reject the agent (revoking the grant per native-delegation §4.2) or absorb the excess cost.

Defense. Iris's monitoring layer rate-limits per-agent attestation volume against the customer's subscription tier. Native delegation grants (companion paper §3.4) carry a time-bound; Iris can include a per-grant attestation cap as part of the application-layer scope predicate (extending the protocol-level scope, which only includes schema set and action set). The combination of per-grant attestation cap + per-billing-period subscription cap limits adversary-customer impact to the customer's own subscription.

Pattern B: base-fee surge exploitation. Adversary agent submits a flood of attestations to drive up b_σ, then continues normal usage at the elevated fee, forcing Iris to pay more per attestation than the pre-committed price curve assumed. This is fee-griefing (§5.4) applied to a sponsor rather than a direct user.

Defense. Iris's pricing model includes a base-fee surge buffer: subscription tiers price against the 90th-percentile observed b_σ over the prior 30-day window, with a clip if observed b_σ exceeds a threshold. If the observed fee climbs beyond the clip, Iris reserves the right to throttle non-subscription-paying volume (e.g., the adversary agent gets de-prioritized while legitimate customers continue at the pre-committed rate). The product-layer pricing discipline handles base-fee surge variance.

Defense (in-protocol part). The per-schema fee market's max-change-per-block ($\xi = 1/8$) caps how quickly an adversary can move b_σ. Iris's pricing buffer needs to absorb at most $\sim$ 12.5% $\times$ (blocks per billing cycle) in worst-case fee climb. At 30-day billing and 12-second block time, that is the theoretical maximum; in practice, fees decay back to equilibrium once the attack stops, so the worst-case is bounded by the attack duration.

Pattern C: routing-fraction exploitation. An adversary registers a schema with $\rho_\sigma = 0.5$ (the maximum) and induces high traffic. They collect $(1-\tau_{\text{burn}})\cdot 0.5$ of every paid base fee. The "attack" here is just a successful schema launch with maximum routing; it is not really an attack, but it is worth noting that schema authors who set $\rho_\sigma = 0.5$ are economically equivalent to chain validators in their cut of fee revenue.

Mitigation. None needed at protocol level; the $\rho_\sigma \leq 0.5$ bound is governance-set and prevents schema authors from extracting more than half of non-burned fees. Beyond that, schema authors who choose high ρ_σ trade lower base-fee-validator-share for higher schema-revenue, which is a legitimate business choice. The cost-to-grind theorem (§5.1) ensures the chain's security floor is unaffected.

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6. Incentive Analysis

6.1 Validator-Side

[**v0.2:** Per-schema fee revenue diversification. Reputation accumulation is schema-agnostic, so validators are not punished for inclusion preferences (within §A.1 bounds).]

6.2 Builder-Side (Block Builders, MEV)

[**v0.2:** Cross-schema bundling, ordering preferences, MEV implications. Defer detailed treatment to a separate paper if needed.]

6.3 Sponsor-Side (Iris and Other Relayers)

[**v0.2:** Sponsored-gas economics. Pre-committed fee curves, retroactive reimbursement, pricing-cap models.]

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7. Implementation in Ligate Chain

7.1 Sovereign SDK Integration Points

[**v0.2:** Per-schema state extension: new `FeeMarketState` field on schemas. Updated runtime hook for base-fee adjustment at epoch boundaries.]

7.2 Recommended v0 Parameters

[**v0.2:** Per-schema T_σ, ξ, $b_\sigma^{\min}$, $b_\sigma^{\max}$ defaults.]

7.3 Migration from Unified Fee Market

[**v0.2:** Bootstrap-period unified fee market; activation of per-schema markets via governance after warmup.]

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8. Comparison with Prior Systems

[**v0.2:** Table comparing this paper, Ethereum EIP-1559, Solana priority fees, Cosmos fee module, Aptos gas pricing across axes: per-resource granularity, target utilization mechanism, burn fraction, sponsor-friendly extensions.]

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9. Limitations and Future Work

9.1 Limitations

[**v0.2:** Empirical validation pending devnet operation; sponsored-gas adversary model is heuristic; multi-resource fee market separation deferred.]

9.2 Future Work

[**v0.2:** Multi-resource fee markets, dynamic schema target adjustment, on-chain fee-market parameter governance.]

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10. Conclusion

[**v0.2:** Per-schema fee markets are the natural unit for attestation-native chains. The mechanism integrates with PoUA reputation cleanly. Combined with the §4.4.2 adaptive τ_{burn} rebase, the chain has both per-schema demand-side price discovery and protocol-level drift response.]

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11. Frequently Asked Questions

[**v0.2:** Q1. Why per-schema and not multi-resource? Q2. How does this interact with EIP-1559 if Ligate later bridges to Ethereum? Q3. What if a schema has near-zero traffic? Q4. Doesn't this just reintroduce the unified-fee-market problems at the schema level? Etc.]

End of working paper v0.1 (outline). Substantive authoring begins with v0.2; see `README.md` and issue #4.